FUNCTORS PRESERVE HOMOTOPIC MORPHISMS OF SIMPLICIAL AND COSIMPLICIAL OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-functorial-homotopy (source lines 5222-5249); the proof environment begins on the second line after it and is its own proof, closing at source line 5256.

Editorial note. The proof is a two-sentence pointer: it reduces to covariant functors by the comparison lemma and refers to the author's remark for the functoriality of the homotopy relation. The card reproduces that as written and carries the remark and the comparison lemma as context, so nothing is silently completed.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: any functor preserves homotopic morphisms of simplicial and cosimplicial objects, in all four variance cases.

Allowed standard prerequisites (editorial).

- Simplicial and cosimplicial objects. The categories of simplicial and of cosimplicial objects in a category, with their morphisms and the notion of homotopy between them.. They are the objects of the four clauses of the statement (source lines 5224-5247).
- Homotopy in terms of the comparison categories. The description of the homotopy relation through the auxiliary categories of the author's remarks, which makes the relation manifestly functorial (the author's remark-homotopy-better and remark-homotopy-cosimplicial-better, retained as context).. It is where the substance of the proof lives (source lines 5254-5255).
- Reduction of a contravariant functor to a covariant one. A contravariant functor may be regarded as a covariant functor between the appropriate pair of categories (the author's lemma-compare-homotopies, retained as context).. It reduces the four cases to the covariant ones (source lines 5252-5253).

Remark 0.1. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects of $\mathcal{C}$. A *homotopy from a to b* is given by morphisms $h_{n,\alpha} : U_n \to V_n$, for $n \geq 0$, $\alpha \in \Delta[1]_n$ satisfying (0.4.1) for all morphisms f of Δ and such that $a_n = h_{n,0:[n]\to[1]}$ and $b_n = h_{n,1:[n]\to[1]}$ for all $n \geq 0$. As in Definition 0.4 we say the morphisms a and

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b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. This new notion is the same as the old one in case finite products exist. We deduce in particular that functors preserve the original notion whenever both categories have finite products.

Remark 0.2. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{C}$. A *homotopy from a to b* is given by morphisms[1] $h_{n,i} : U_n \to V_n$, for $n \geq 0$, $i = 0, \ldots, n+1$ satisfying the relations of Lemma 0.6. As in Definition 0.5 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. Since the lemma says that the newer notion is the same as the old one in case finite coproduct exist, we deduce in particular that functors preserve the original notion whenever both categories have finite coproducts.

Lemma 0.3. *Let $\mathcal{C}$ be a category. Suppose that U and V are two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of cosimplicial objects. Recall that U, V correspond to simplicial objects U', V' of $\mathcal{C}^{opp}$. Moreover a, b correspond to morphisms $a', b' : V' \to U'$. The following are equivalent*

1. *There exists a homotopy $h = \{h_{n,\alpha}\}$ from a to b as in Remark 0.1.*
2. *There exists a homotopy $h = \{h_{n,i}\}$ from a' to b' as in Remark 0.2.*

Thus a is homotopic to b as in Remark 0.1 if and only if a' is homotopic to b' as in Remark 0.2.

Proof. In case $\mathcal{C}$ has finite products, then $\mathcal{C}^{opp}$ has finite coproducts and we may use Definitions 0.4 and 0.5 instead of Remarks 0.1 and 0.2. In this case $h : U \to \mathrm{Hom}(\Delta[1], V)$ is the same as a morphism $h' : \mathrm{Hom}(\Delta[1], V)' \to U'$. Since products and coproducts get switched too, it is immediate that $(\mathrm{Hom}(\Delta[1], V))' = V' \times \Delta[1]$. Moreover, the "primed" version of the morphisms $e_0, e_1 : \mathrm{Hom}(\Delta[1], V) \to V$ are the morphisms $e_0, e_1 : V' \to \Delta[1] \times V$. Thus $e_0 \circ h = a$ translates into $h' \circ e_0 = a'$ and similarly $e_1 \circ h = b$ translates into $h' \circ e_1 = b'$. This proves the lemma in this case.

In the general case, one needs to translate the relations given by (0.4.1) into the relations given in Lemma 0.6. We omit the details.

The final assertion is formal from the equivalence of (1) and (2). □

Definition 0.4. Let $\mathcal{C}$ be a category having finite products. Let U and V be two cosimplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms of cosimplicial objects of $\mathcal{C}$.

[1] In the literature, often the maps $h_{n+1,i} \circ s_i : U_n \to V_{n+1}$ are used instead of the maps $h_{n,i}$. Of course the relations these maps satisfy are different from the ones in Lemma 0.6.

(1) We say a morphism

$$h : U \longrightarrow \mathrm{Hom}(\Delta[1], V)$$

such that $a = e_0 \circ h$ and $b = e_1 \circ h$ is a *homotopy from a to b*.
(2) We say a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence $a = a_0, a_1, \ldots, a_n = b$ of morphisms from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_i to a_{i-1} or there exists a homotopy from a_{i-1} to a_i.

$$h_{m,\alpha} \circ U(f) = V(f) \circ h_{n,\alpha\circ f} \tag{0.4.1}$$

Definition 0.5. Let $\mathcal{C}$ be a category having finite coproducts. Suppose that U and V are two simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms.

(1) We say a morphism

$$h : U \times \Delta[1] \longrightarrow V$$

is a *homotopy from a to b* if $a = h \circ e_0$ and $b = h \circ e_1$.
(2) We say the morphisms a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}.

Lemma 0.6. *In the situation above, we have the following relations:*

(1) *We have $h_{n,0} = b_n$ and $h_{n,n+1} = a_n$.*
(2) *We have $d^n_j \circ h_{n,i} = h_{n-1,i-1} \circ d^n_j$ for $i > j$.*
(3) *We have $d^n_j \circ h_{n,i} = h_{n-1,i} \circ d^n_j$ for $i \leq j$.*
(4) *We have $s^n_j \circ h_{n,i} = h_{n+1,i+1} \circ s^n_j$ for $i > j$.*
(5) *We have $s^n_j \circ h_{n,i} = h_{n+1,i} \circ s^n_j$ for $i \leq j$.*

Conversely, given a system of maps $h_{n,i}$ satisfying the properties listed above, then these define a morphism h which is a homotopy from a to b.

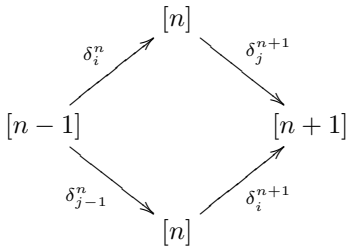
Proof. Omitted. You can prove the last statement using the fact, see Lemma 0.7 that to give a morphism of simplicial objects is the same as giving a sequence of morphisms h_n commuting with all d^n_j and s^n_j. □

Lemma 0.7. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.8 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

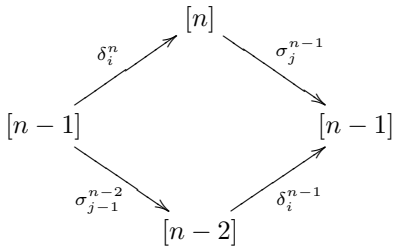
Lemma 0.8. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If* $0 \leq i < j \leq n+1$, *then* $\delta_j^{n+1} \circ \delta_i^n = \delta_i^{n+1} \circ \delta_{j-1}^n$. *In other words the diagram*



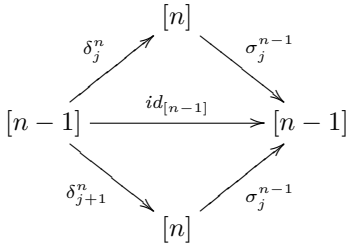
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(2) *If* $0 \leq i < j \leq n-1$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_i^{n-1} \circ \sigma_{j-1}^{n-2}$. *In other words the diagram*



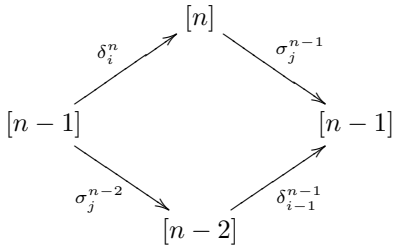
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(3) *If* $0 \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \delta_j^n = id_{[n-1]}$ *and* $\sigma_j^{n-1} \circ \delta_{j+1}^n = id_{[n-1]}$. *In other words the diagram*



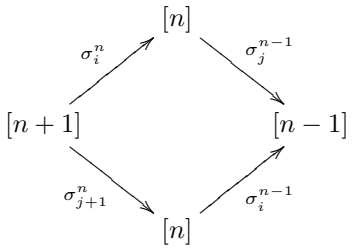
commutes.

(4) *If* $0 < j+1 < i \leq n$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. *In other words the diagram*



commutes.

(5) *If* $0 \leq i \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. *In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.9. *Let* $\mathcal{C}, \mathcal{C}', \mathcal{D}, \mathcal{D}'$ *be categories. With terminology as in Remarks 0.1 and 0.2.*

(1) *Let* $a, b : U \to V$ *be morphisms of simplicial objects of* $\mathcal{D}$. *Let* $F : \mathcal{D} \to \mathcal{D}'$ *be a covariant functor. If* a *and* b *are homotopic, then* $F(a)$, $F(b)$ *are homotopic morphisms* $F(U) \to F(V)$ *of simplicial objects.*
(2) *Let* $a, b : U \to V$ *be morphisms of cosimplicial objects of* $\mathcal{C}$. *Let* $F : \mathcal{C} \to \mathcal{C}'$ *be a covariant functor. If* a *and* b *are homotopic, then* $F(a)$, $F(b)$ *are homotopic morphisms* $F(U) \to F(V)$ *of cosimplicial objects.*
(3) *Let* $a, b : U \to V$ *be morphisms of simplicial objects of* $\mathcal{D}$. *Let* $F : \mathcal{D} \to \mathcal{C}$ *be a contravariant functor. If* a *and* b *are homotopic, then* $F(a)$, $F(b)$ *are homotopic morphisms* $F(V) \to F(U)$ *of cosimplicial objects.*
(4) *Let* $a, b : U \to V$ *be morphisms of cosimplicial objects of* $\mathcal{C}$. *Let* $F : \mathcal{C} \to \mathcal{D}$ *be a contravariant functor. If* a *and* b *are homotopic, then* $F(a)$, $F(b)$ *are homotopic morphisms* $F(V) \to F(U)$ *of simplicial objects.*

Proof. By Lemma 0.3 above, we can turn F into a covariant functor between a pair of categories, and we have to show that the functor preserves homotopic pairs of maps. This is explained in Remark 0.2. □